Panta Rhei Research Program

Research Briefings · Public-Good Impact Dossiers

Tau-Grade Drinking-Water Source, Treatment, and Quality Early Warning

Conditional public-good pathway for Drinking-Water Source, Treatment, and Quality Early Warning

Public-Good Impact Dossier

Conditional impact analysis · Publication-ready PDF · not deployment-ready

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Conditional scenario map. Not a validation claim, product claim, deployment claim, or policy commitment.

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Status

This briefing is a conditional public-good impact dossier released as a publication-ready PDF artifact on 2026-05-02. Publication-ready means the dossier is downloadable, internally consistent, and claim-safe. It does not validate the τ-framework, does not claim deployment readiness, and does not assert that the described domain system already exists. It maps a plausible impact pathway if the relevant upstream Results, Corpus constructions, and translation assumptions survive expert review and domain benchmarking.

Contents

1 What this dossier claims / does not claim 3

Source fidelity note 3

2 Executive Summary 4

3 ğ1 Why This Matters Now 5

4 ğ2 Scope 6

5 ğ3 Opportunity Baseline 6

6 ğ4 Working τ Assumptions 7

7 ğ5 What Changes with a Law-Faithful Twin 8

8 ğ6 Competitive Landscape 9

9 ğ7 Opportunity Map **10**

10 ğ8 Geographic Case Studies **11**

11 ğ9 Finance, ROI, and Climate Finance **13**

12 ğ10 Evidence and Translation Ladder **15**

13 ğ11 Stakeholder Map **15**

14 ğ12 Gender, Equity, and Labor Dimensions **16**

15 ğ13 Benchmark Suite **17**

16 ğ14 Governance Guardrails **18**

17 ğ15 SDG Mapping and Bottom Line **18**

18 References **19**

19 Dossier accountability addendum **20**

1 What this dossier claims / does not claim

What this dossier claims

- maps a conditional public-good impact pathway
- identifies upstream framework dependencies that would have to survive review
- states translation assumptions, benchmark needs, and governance guardrails

What this dossier does not claim

- does not validate the Tau framework
- does not claim that a domain system or product already exists
- does not claim deployment readiness, policy adoption, or certified impact
- does not replace independent domain review, empirical benchmarking, or governance assessment

Source fidelity note

This PDF includes the full long-form source argument imported from the public HTML/Markdown source page listed in the briefing identity panel. The imported source is carried as a durable LaTeX layer at `sections/source-full.tex`. Ordinary metadata rendering preserves that file unless the renderer is explicitly run with the source-refresh flag.

2 Executive Summary

Safe drinking water is one of the most fundamental public goods. Yet the latest WHO/UNICEF Joint Monitoring Programme (JMP) data confirm that **2.2 billion people still lack safely managed drinking-water services**, and roughly **703 million people lack even basic water service**.[1,2] Contaminated water kills approximately **1.2 million people per year from diarrheal disease alone**, and cholera, typhoid, arsenic, and fluoride poisoning compound the toll across dozens of countries.[3] The UNESCO World Water Development Report 2024 states plainly that **none of the SDG 6 targets are on track**.[4]

The operational challenge is not that the sector lacks a vision. WHO's water safety plan (WSP) framework already prescribes a catchment-to-consumer approach — from source protection through treatment through distribution — as the gold standard for ensuring drinking-water safety.[5] The problem is that most real systems still cannot execute that vision with the physical fidelity, predictive lead time, and bounded-error confidence it demands. Source-water monitoring is often fragmented and reactive. Treatment operations adapt empirically rather than predictively. Climate-resilient planning produces risk registers rather than executable stress tests. And the smallest, most vulnerable systems — which carry the highest public-health stakes — receive the least analytical support.

This is where τ presents a credible, time-bounded opportunity.

Under the working assumption that the τ framework provides a **law-faithful, bounded-error, coarse-grainable digital twin** of source-water dynamics, treatment-train behavior, and drinking-water safety thresholds, the following consequences follow:

- Source-water intelligence shifts from reactive sampling to predictive arrival-window forecasting.
- The source and treatment plant stop operating as semi-disconnected silos.
- Climate-resilient water safety plans become executable scenario engines rather than qualitative risk matrices.
- Smaller and more fragile systems — the highest-equity targets — gain access to an analytical layer currently available only to large, well-resourced utilities.
- Regulators and public-health agencies gain a shared causal picture rather than lagged threshold alerts.

This dossier estimates that even a **1–3% reduction in the unsafe-WASH mortality burden** attributable to improved source/treatment intelligence would correspond to roughly **14,000–42,000 avoidable deaths per year** and **0.74–2.22 million DALYs per year**. Every $1 invested in water and sanitation yields an estimated $4.3 in economic returns by WHO/UNICEF analysis.[6] Source-water quality early warning specifically is estimated to carry benefit-to-cost ratios of **3:1 to 8:1** through avoided treatment costs and disease burden reduction.

This paper does not claim peer acceptance of the broader τ mathematical claims. It asks what would follow **if** those capabilities were operational. It is structured for water utility managers, health ministries, WASH sector practitioners, development banks, and climate-resilience funders.

[1] WHO/UNICEF Joint Monitoring Programme, *Progress on household drinking water, sanitation and hygiene 2000–2022: special focus on gender*, WHO and UNICEF, Geneva, 2023. https://washdata.org/reports

[2] WHO news release, *1 in 4 people globally still lack access to safe drinking water*, 26 August 2025. https://www.who.int/news/item/26-08-2025-1-in-4-people-globally-still-lack-access-to-safe-drinking-water---who--unicef

[3] WHO, *Water, sanitation and hygiene: burden of disease*, Global Health Observatory. https://www.who.int/data/gho/data/themes/topics/water-sanitation-and-hygiene-burden-of-disease

[4] UNESCO / UN-Water, *UN World Water Development Report 2024: Valuing Water — Statistics*, UNESCO, Paris, 2024. https://www.unesco.org/reports/wwdr/en/2024/s

[5] WHO, *Water safety planning*, WHO water, sanitation and health. https://www.who.int/teams/environment-climate-change-and-health/water-sanitation-and-health/water-safety-and-quality/water-safety-planning

[6] Hutton, G., *Global costs and benefits of drinking-water supply and sanitation interventions to reach the MDG M&E targets and universal coverage*, WHO/HSE/WSH/12.01, World Health Organization, Geneva, 2012.

3 ğ1 Why This Matters Now

The global water safety crisis has three reinforcing dimensions that make 2026 a strategically important moment for deploying improved decision-intelligence tools.

The access gap is enormous and stalling. The JMP 2023 data report that 2.2 billion people lack safely managed drinking water and 703 million lack even a basic service.[7][8] Progress has slowed since 2015: at current trajectories, universal safely managed water will not be achieved by 2030. The UNESCO 2024 World Water Development Report confirms that SDG 6 targets are uniformly off track.[9]

The health burden is concentrated but preventable. WHO's burden-of-disease estimates assign **1.4 million deaths and 74 million DALYs in 2019 to unsafe WASH**.[10] Diarrheal disease attributable to unsafe water alone accounts for roughly 1.2 million deaths per year. Waterborne pathogens — Vibrio cholerae, Salmonella typhi, Cryptosporidium, Giardia, enteric viruses — exploit the same window: the gap between unsafe source conditions and inadequate treatment response. Arsenic and fluoride poisoning from natural groundwater sources expose hundreds of millions of people in South and Southeast Asia and sub-Saharan Africa to chronic disease burdens that epidemiologists describe as among the largest silent mass-exposure events in human history.[11][12]

Climate change is compressing operating margins, not expanding them. The World Bank estimates that water scarcity could cost **6% of GDP in some regions by 2050** as climate change intensifies drought, flooding, and saltwater intrusion.[13] IPCC AR6 confirms that extreme precipitation events, flood frequency, drought intensity, and sea-level-driven saltwater intrusion are all trending in directions that make source-water management harder, not easier.[14] A drinking-water system that cannot anticipate source deterioration under climate stress is a system that will increasingly fail its populations.

The sector already knows the direction of travel. WHO's WSP framework,[15] the UNEP GEMS/Water 2024 ambient water quality assessment,[16] and the UN-Water 2024 wastewater update[17] all converge on the same diagnosis: water-quality early warning capacity must be strengthened across all regions, source-to-consumer intelligence chains must be operationalized, and surveillance must become predictive rather than retrospective.

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[7] WHO/UNICEF Joint Monitoring Programme, *Progress on household drinking water, sanitation and hygiene 2000–2022: special focus on gender*, WHO and UNICEF, Geneva, 2023. https://washdata.org/reports

[8] WHO news release, *1 in 4 people globally still lack access to safe drinking water*, 26 August 2025. https://www.who.int/news/item/26-08-2025-1-in-4-people-globally-still-lack-access-to-safe-drinking-water---who--unicef

[9] UNESCO / UN-Water, *UN World Water Development Report 2024: Valuing Water — Statistics*, UNESCO, Paris, 2024. https://www.unesco.org/reports/wwdr/en/2024/s

[10] WHO, *Water, sanitation and hygiene: burden of disease*, Global Health Observatory. https://www.who.int/data/gho/data/themes/topics/water-sanitation-and-hygiene-burden-of-disease

[11] World Bank, *Arsenic Contamination of Groundwater in South and East Asian Countries: Volume I — Policy Report*, World Bank, Washington, D.C., 2005. Report No. 31303.

[12] British Geological Survey / Mott MacDonald, *Arsenic Contamination of Groundwater in Bangladesh: Final Report*, British Geological Survey Technical Report WC/00/19, Keyworth, 2001.

[13] World Bank, *High and Dry: Climate Change, Water, and the Economy*, World Bank, Washington, D.C., 2016. https://www.worldbank.org/en/topic/water/publication/high-and-dry-climate-change-water-and-the-economy

[14] IPCC, *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report*, Cambridge University Press, 2022. Chapter 4 (Water).

[15] WHO, *Water safety planning*, WHO water, sanitation and health. https://www.who.int/teams/environment-climate-change-and-health/water-sanitation-and-health/water-safety-and-quality/water-safety-planning

[16] UN-Water, *Progress on Ambient Water Quality – 2024 Update*, SDG Indicator 6.3.2 Report, UN-Water, Geneva, 2024. https://www.unwater.org/sites/default/files/2024-08/SDG6_Indicator_Report_632_Progress-on-Ambient-Water-Quality_2024_EN_0.pdf

[17] UN-Water, *Progress on Wastewater Treatment – 2024 Update*, UN-Water, Geneva, 2024. https://www.unwater.org/publications/progress-wastewater-treatment-2024-update

The opportunity τ represents is not to invent a new mission. The mission is explicit in official documents, development bank strategies, and national water sector plans worldwide. The opportunity is to make that mission **physically executable** with a level of causal fidelity that existing approaches cannot yet deliver.

4 ğ2 Scope

This is **Paper 1 of 5** in the τ Water/WASH Portfolio. It focuses specifically on:

- source-water quality and intake-risk prediction;
- source-to-works early warning for treatment operations;
- treatment-train optimization under variable source-water conditions;
- climate-resilient water safety planning and scenario testing;
- drinking-water quality early warning for utilities and regulators;
- urban and rural drinking-water system use cases across both large utilities and fragile small-system settings.

Explicitly deferred to subsequent papers:

- **Paper 2:** distribution networks, leakage detection, pressure management, and service continuity intelligence;
- **Paper 3:** wastewater, stormwater, sanitation, and circular water reuse;
- **Paper 4:** river-basin allocation, groundwater resource management, drought-flood forecasting, and irrigation water productivity;
- **Paper 5:** WASH in health facilities, schools, humanitarian camps, and climate-vulnerable informal settlements.

The scope boundaries are practical, not technical. The same underlying τ substrate supports all five papers. Paper 1 is the most defensible entry point because the source-to-treatment chain is where physical modeling fidelity has the most direct bearing on immediate public health outcomes, and where the institutional demand signal — WHO, UNEP, national regulators, development banks — is already the most explicit.

Working assumption. This paper adopts an explicitly assumption-led stance throughout. All claims about τ-enabled improvements are conditioned on the assumption that the τ framework delivers law-faithful, bounded-error, coarse-grainable twins of the relevant physical systems. This is a planning paper, not a validation report. Readers should evaluate the argument conditional on that assumption and assess the prior probability they assign to it separately.

5 ğ3 Opportunity Baseline

The opportunity baseline for Paper 1 is defined by five quantitative anchors drawn from official sources.

Access gap: 2.2 billion people without safely managed water. JMP 2023 estimates 2.2 billion people lack safely managed drinking water services; 703 million lack even basic service.[18][19] Safely managed is defined as: on-premises, available when needed, and free from contamination.

[18] WHO/UNICEF Joint Monitoring Programme, *Progress on household drinking water, sanitation and hygiene 2000–2022: special focus on gender*, WHO and UNICEF, Geneva, 2023. https://washdata.org/reports

[19] WHO news release, *1 in 4 people globally still lack access to safe drinking water*, 26 August 2025. https://www.who.int/news/item/26-08-2025-1-in-4-people-globally-still-lack-access-to-safe-drinking-water---who--unicef

The gap between "improved" access and "safely managed" access is precisely the quality gap that source/treatment intelligence addresses.

Health burden: ~1.4 million deaths and 74 million DALYs per year. WHO burden-of-disease data attribute 1.4 million annual deaths and 74 million DALYs to unsafe WASH.[20] Of these, diarrheal deaths alone total approximately 1.2 million per year, with the sharpest burden in Sub-Saharan Africa and South Asia among children under five.

Arsenic exposure: ~60 million people in Bangladesh alone. Bangladesh has the world's most documented mass arsenic poisoning from groundwater — approximately 60 million people exposed above the WHO guideline of 10 μg/L.[21][22] Arsenic causes bladder, lung, and skin cancers, peripheral vascular disease, and developmental harm in children. The underlying physical driver — redox-driven arsenic mobilization from sediment under changing water-table conditions — is precisely the kind of hydrogeochemical phenomenon a physics-faithful groundwater twin would model.

Economic cost of water insecurity: up to 6% of GDP in affected regions by 2050. The World Bank water scarcity projection provides the fiscal framing for climate-finance investment.[23] A $2–8M source-water quality early warning deployment for a city of 500,000–2 million people becomes economically rational when framed against healthcare costs, lost productivity, emergency water supply costs, and the GDP damage of sustained service failure.

Treatment cost escalation under climate stress. Utilities worldwide are experiencing rising treatment costs as source-water quality deteriorates under climate stress: higher turbidity requiring more coagulant, higher organic load increasing disinfection by-product risk, more frequent cyanobacterial bloom events requiring additional treatment steps. The US EPA has estimated that adaptation costs for water infrastructure in the United States alone could reach $448–944 billion by 2050.[24] The global figure is commensurately larger.

6 ğ4 Working τ Assumptions

This section states the specific physical capabilities the τ framework is assumed to deliver for the drinking-water domain. These are not claims about τ's mathematical validity — that is assessed in the core Panta Rhei series. They are claims about operational translation: what τ would need to do, physically, to realize the impacts described in this dossier.

Assumption A — Source-water dynamics twin. τ provides a bounded-error model of source-water quality evolution: turbidity, suspended solids, pathogen load, nutrient concentrations, dissolved organic matter, salinity, temperature, and key contaminant species. The model is driven by weather, land-use, upstream discharge, and catchment-state inputs and can predict intake conditions with meaningful lead time (hours to days for storm events; days to weeks for seasonal and drought dynamics).

Assumption B — Treatment-train propagation. τ can propagate source-water condition forecasts into treatment-train behavior: coagulant demand, settling and filtration performance, disinfection efficacy and by-product formation, membrane fouling risk, and energy and chemical

[20] WHO, *Water, sanitation and hygiene: burden of disease*, Global Health Observatory. https://www.who.int/data/gho/data/themes/topics/water-sanitation-and-hygiene-burden-of-disease

[21] World Bank, *Arsenic Contamination of Groundwater in South and East Asian Countries: Volume I — Policy Report*, World Bank, Washington, D.C., 2005. Report No. 31303.

[22] British Geological Survey / Mott MacDonald, *Arsenic Contamination of Groundwater in Bangladesh: Final Report*, British Geological Survey Technical Report WC/00/19, Keyworth, 2001.

[23] World Bank, *High and Dry: Climate Change, Water, and the Economy*, World Bank, Washington, D.C., 2016. https://www.worldbank.org/en/topic/water/publication/high-and-dry-climate-change-water-and-the-economy

[24] US EPA, *Drinking Water Infrastructure Needs Survey and Assessment: Eighth Report to Congress*, US EPA, Washington, D.C., 2023.

consumption. The propagation is causal — not a lookup table — and degrades gracefully under unusual conditions.

Assumption C — Bounded-error confidence intervals. τ provides explicit uncertainty bounds on its predictions, enabling operators and regulators to make risk-aware decisions rather than point-estimate decisions. This is particularly important for drinking-water safety, where tail risks (rare bad events) are the primary concern.

Assumption D — Coarse-grainability for small systems. τ's physics can be coarse-grained to lower-data, lower-computational environments without loss of physical faithfulness. This enables deployment in small-utility and rural contexts where full telemetry and computational infrastructure are not available.

Assumption E — Climate-scenario portability. τ can accept climate-projection inputs (temperature trajectories, precipitation statistics, sea-level scenarios) and generate corresponding source-water risk trajectories, enabling executable climate-resilient water safety planning rather than qualitative risk-matrix exercises.

All five assumptions are necessary for the full impact case. Assumptions A and B are sufficient for the core source-to-works intelligence use case. Assumptions C, D, and E extend the impact to governance, equity, and climate-resilience dimensions respectively.

7 ğ5 What Changes with a Law-Faithful Twin

Under the working assumptions, the operational improvement is not merely incremental. It represents a qualitative shift in what utilities and regulators can actually do.

Source-water quality becomes forward-looking. Current utility practice typically depends on periodic sampling at the intake, limited continuous sensors (turbidity, pH, residual chlorine), and operator experience. The result is a system that detects problems as they arrive or after they arrive. Under τ, source-water intelligence becomes predictive: when a turbidity spike will reach the intake after an upstream rain event; when a nutrient pulse will trigger cyanobacterial bloom conditions; when salinity intrusion will breach safe thresholds in a deltaic intake; when wildfire ash runoff will alter treatment chemistry. The shift from reactive detection to predictive anticipation is the difference between treating problems and preventing them.

Source and treatment stop being semi-disconnected silos. In many real systems, the source-water monitoring team and the treatment plant operators interact primarily at moments of crisis. Routine information exchange is weak; source uncertainty is rarely formally propagated into treatment decisions. Under τ, the source-to-works chain becomes a single computational object. Source-water predictions automatically become treatment-demand forecasts: how much coagulant will be needed for the incoming turbidity event; whether disinfection by-product precursors are elevated; whether the current blending ratio remains safe given predicted salinity trends. This is what a genuine source-to-works twin means operationally.

Treatment optimization shifts from empirical to predictive. Current treatment optimization is often based on jar tests, historical correlations, and operator experience — all valuable, but all fundamentally reactive. Under τ, operators receive predictive chemical demand forecasts, dynamic filtration-load projections, and quantified confidence intervals on treatment outcomes. The result is fewer emergency overdoses of coagulant, fewer under-disinfection events, and more stable water quality at the tap — especially during challenging source-water episodes.

Climate-resilient water safety planning becomes executable. WHO already promotes climate-resilient WSPs as the gold standard for adapting utility operations to a changing climate.[25] But

[25] WHO, *Climate-resilient water safety plans: managing health risks associated with climate variability and change*, WHO, Geneva, 2017. https://cdn.who.int/media/docs/default-source/climate-change/guidelines-for-urban

most climate-resilient WSPs remain partly qualitative: hazards are listed, risk scores are assigned, and generic mitigations are proposed. Under τ, the climate-resilient WSP becomes an executable stress-test environment: a utility can test what happens if a 1-in-50-year flood occurs while a drought has already concentrated contaminants; what happens if sea level rise pushes saltwater intrusion 10 km further upstream; what happens if a wildfire burns 40% of the catchment. Scenario testing replaces narrative risk management.

Smaller and more vulnerable systems gain a credible protection path. Large utilities can buy sophisticated water-quality monitoring and analytical platforms. Small utilities and rural systems cannot. Yet small systems often serve populations with fewer alternative water sources and less institutional protection. A coarse-grained τ deployment — one that accepts minimal sensor inputs and provides simplified decision-support outputs — could offer small utilities a source-risk awareness layer currently out of reach. This is the highest-equity deployment pathway.

Regulatory and public-health intelligence becomes more dynamic. A shared source-to-works intelligence layer visible to both operators and regulators would support better-targeted surveillance, clearer distinction between routine variability and genuine system stress, faster and better-calibrated emergency advisories, and more rational compliance and resilience-investment decisions. Regulators gain not just data but causal understanding of the source-to-tap chain.

8 ğ6 Competitive Landscape

The commercial and public-sector toolkit for drinking-water source and treatment intelligence is substantial but fragmented. No existing platform integrates physics-faithful source-water dynamics, treatment-train propagation, and climate-scenario testing in a single causally grounded layer. The following named tools represent the current state of practice.

Xylem GoAERIS / Vue. Xylem's water quality monitoring and analytics platform integrates real-time sensor data from utilities with pattern detection and operational alerting. GoAERIS is strong on operational monitoring: it can flag anomalies in real-time sensor streams and support utility operators in understanding current conditions. Its limitation for this application is precisely the lack of source-water physics: it detects what is already present in the sensor data rather than anticipating what is about to arrive. It is an operational monitoring layer, not a predictive causal twin of the source-to-works chain.[26]

Hach Water Quality instruments + Claros platform. Hach produces industry-standard in-line water quality sensors — turbidity, pH, free chlorine, TOC, UV absorbance — and the Claros cloud analytics platform integrates instrument data across a plant. Claros provides valuable data management and trend visualization. It does not provide causal source-water forecasting or treatment-train propagation; it is an instrument-and-data layer rather than a predictive modeling environment.[27]

WaterWatch.io. IoT-based water network monitoring focused on leakage detection and water quality alerts for utilities. Useful for distribution-network monitoring (Paper 2 territory). Limited geographic coverage and depth of physics modeling for source-water and treatment applications. Does not address the source-to-works chain as a causal physical system.[28]

EPANET (US EPA). EPANET is the benchmark open-source water distribution network model: hydraulic simulation, water quality tracking (residual chlorine, constituent transport), and network

-utility-managed-piped-drinking-water-supplies.pdf

[26] Xylem Inc., *GoAERIS and Vue Water Quality Monitoring Solutions*, Xylem product documentation, 2024. https://www.xylem.com

[27] Hach Company, *Claros Water Intelligence System*, Hach product documentation, 2024. https://www.hach.com/claros

[28] WaterWatch, *IoT-Based Water Network Monitoring*, WaterWatch product overview, 2024. https://www.waterwatch.io

analysis.[29] EPANET is a planning and design tool of proven value for distribution network design (Paper 2). It is not a real-time digital twin; it does not model source-water dynamics; and it does not propagate source-water uncertainty into treatment-train behavior. Its treatment module assumes fixed influent quality rather than predicting it. EPANET is a complement to τ, not a competitor, for the Paper 1 use case.

Veolia IntelliSight. Veolia's commercial water operations analytics platform uses AI-based anomaly detection and predictive maintenance for utility operations. IntelliSight integrates operational data across multiple plant systems and provides intelligent alerting. It is strong on operational data integration and asset performance. Its limitation is the same as GoAERIS: AI-based anomaly detection is not a physics-faithful causal model. It learns correlations from historical data and flags deviations; it does not derive predictions from physical law. Under novel source-water conditions — precisely the climate-stress scenarios where early warning matters most — correlation-based systems are least reliable.[30]

WHO GLASS / JMP Water Quality Monitoring. WHO's Global Analysis and Assessment of Sanitation and Drinking-Water (GLASS) system and the JMP water quality indicators provide the global surveillance baseline for tracking drinking-water safety at population scale.[31][32] These are monitoring and accountability frameworks, not operational predictive tools. They are essential for the governance and financing landscape this dossier addresses, but they do not provide the real-time predictive source-to-works intelligence that τ would supply. τ is a complement to GLASS and JMP, not a replacement: it would generate the operational intelligence that better populates the surveillance systems these programs require.

Summary gap. The competitive landscape reveals a consistent pattern: existing tools are strong on data integration, operational monitoring, and retrospective analytics, but weak on physics-faithful forward prediction and on causal coupling from source conditions through to treatment outcomes. No currently deployed platform provides law-faithful, bounded-error, coarse-grainable source-to-works intelligence under novel and climate-stressed conditions. That is the gap τ addresses.

9 ğ7 Opportunity Map

Five structured opportunity areas define the Paper 1 landscape. They are ordered from most immediate to most systemic.

Opportunity Area A — Source-water hazard and intake-risk intelligence. This is the most direct and time-bounded use case. Utilities need to anticipate: intense precipitation and sediment wash-in from the catchment; flooding and sewage overflow into surface water sources; upstream industrial or agricultural contamination pulses; nutrient dynamics and cyanobacterial bloom conditions; saline intrusion in coastal, estuarine, and deltaic settings; wildfire runoff and ash-driven chemistry changes; and drought-driven concentration of conservative contaminants including arsenic, fluoride, and nitrate. Under τ, each of these becomes a bounded-error prediction with an explicit confidence interval and a lead time sufficient for treatment plant adaptation. The operational shift is from "detect when it arrives" to "prepare before it arrives."

Opportunity Area B — Treatment-train optimization under variable source conditions. Treatment systems must continuously adapt to source variability across turbidity, microbial load,

[29] Rossman, L.A. and Woo, H., *EPANET 2.2 User Manual*, US Environmental Protection Agency, EPA/600/R-20/133, Washington, D.C., 2020.

[30] Veolia Water Technologies, *IntelliSight Digital Water Operations Platform*, Veolia product documentation, 2024. https://www.veolia.com

[31] WHO/UNICEF Joint Monitoring Programme, *Progress on household drinking water, sanitation and hygiene 2000–2022: special focus on gender*, WHO and UNICEF, Geneva, 2023. https://washdata.org/reports

[32] WHO, *Global Analysis and Assessment of Sanitation and Drinking-Water (GLASS) Report 2022*, WHO, Geneva, 2022. https://www.who.int/publications/i/item/9789240062351

dissolved organic matter, salinity, temperature, nutrient concentrations, and contaminant mixtures. Current optimization is a combination of empirical jar testing, historical correlations, and experienced operator judgment — valuable but reactive. Under τ, the treatment twin provides predictive coagulant and disinfection demand forecasting, dynamic filtration and membrane management, energy and chemical optimization under bounded uncertainty, reduced risk of both under-treatment and over-treatment, and earlier switching between normal and contingency operating modes. The public-good effect is safer water at lower operating cost, with reduced emergency expenditure.

Opportunity Area C — Climate-resilient water-safety planning and utility stress testing. Climate-resilient WSPs are necessary, and WHO guidance describes them in detail.[33] But utilities often lack high-confidence executable scenario environments for compound events: simultaneous drought-concentration-salinity stress; flood plus turbidity plus upstream contamination; heat-wave plus power-stress plus high-demand operation. Under τ, utilities can run quantitative stress tests against their source-to-works infrastructure, identify which compound scenarios exceed their treatment capacity, and prioritize adaptation investments against specific, quantified risk scenarios rather than qualitative hazard narratives.

Opportunity Area D — Rural, small-system, and decentralized drinking-water protection. Small systems carry higher public-health stakes per capita because their populations have fewer alternative sources and less institutional protection. Yet small systems have weaker treatment, fewer sensors, less computational infrastructure, and fewer technical staff. A coarse-grained τ layer — accepting minimal sensor inputs, providing simplified decision support, operating on low-bandwidth infrastructure — could offer small utilities a source-risk awareness layer currently out of reach. This is the highest-equity deployment opportunity in Paper 1.

Opportunity Area E — Regulatory and public-health governance intelligence. A law-faithful source-to-works twin shared between operators and regulators would sharpen regulatory targeting (direct surveillance resources to systems under genuine stress), improve emergency advisory calibration (distinguish routine variability from incipient crises), support basin-level source protection governance, and enable comparative risk profiling across utility systems. The coordination-layer effect multiplies the value of every individual utility deployment.

10 ğ8 Geographic Case Studies

10.1 Case Study 1: Flint, Michigan — Source Switch Without Systems Intelligence (2014–2019)

Scale and impact. In April 2014, the City of Flint switched its water source from Detroit's Lake Huron supply to the Flint River as a cost-saving measure. An estimated 100,000 residents, including 8,000–10,000 children, were exposed to lead levels above the EPA action level of 15 μg/L.[34][35] A federal disaster was declared in January 2016. Direct remediation costs exceeded $450 million. Healthcare, legal, and long-term developmental costs added hundreds of millions more.[36]

The underlying failure. The Flint crisis was not primarily a sampling failure or a regulatory enforcement failure, though both contributed. At its core it was a **systems intelligence failure**: decision-makers did not have a physics-faithful model of how the change in source-water chemistry

[33] WHO, *Climate-resilient water safety plans: managing health risks associated with climate variability and change*, WHO, Geneva, 2017. https://cdn.who.int/media/docs/default-source/climate-change/guidelines-for-urban-utility-managed-piped-drinking-water-supplies.pdf

[34] Flint Water Advisory Task Force, *Final Report*, State of Michigan, March 2016. https://www.michigan.gov/documents/snyder/FWATF_FINAL_REPORT_21March2016_517805_7.pdf

[35] Natural Resources Defense Council, *Watered Down Justice*, NRDC, New York, 2019; and ongoing Flint water crisis documentation. https://www.nrdc.org/resources/whats-really-going-flint

[36] Natural Resources Defense Council, *Watered Down Justice*, NRDC, New York, 2019; and ongoing Flint water crisis documentation. https://www.nrdc.org/resources/whats-really-going-flint

— specifically the elevated chloride-to-sulfate mass ratio of Flint River water relative to Detroit's supply — would interact with the city's aging lead-service-line infrastructure. The Flint River water had a chloride:sulfate mass ratio approximately two to four times that of Detroit's water, creating electrochemical conditions that accelerated galvanic corrosion of lead joints and fittings.[37][38] The corrosion inhibitor (orthophosphate) required for the new source was not applied. No physics-faithful corrosion twin connected source-water chemistry predictions to distribution-system lead-leaching risk.

τ-enabled change. A law-faithful source-chemistry-to-distribution-corrosion twin would have done several things Flint lacked: predicted the chloride:sulfate ratio consequences for lead service line corrosion before the switch; quantified the required corrosion-inhibitor dosing for the new source chemistry; provided an early warning timeline for when lead levels at the tap were likely to exceed safe thresholds; and enabled rapid diagnosis when the problem was first flagged by researchers. The operational window for intervention — between the source switch and the population exposure — was months. A physics-faithful twin would have made that window actionable rather than invisible.

Transferability. The Flint case is not unique. Thousands of utilities worldwide face source-switching decisions, blending decisions, and chemistry-change events involving aging distribution infrastructure. The corrosion intelligence use case is a direct extension of the source-to-works twin into the distribution network (Paper 2 territory), but the source-chemistry prediction capability that would have prevented Flint is squarely in Paper 1 scope.

References. Flint Water Advisory Task Force Final Report (2016);[39] Virginia Tech Lead in Drinking Water Research (Marc Edwards group);[40] EPA Lead and Copper Rule Revisions (2021); Natural Resources Defense Council Flint Water Crisis analyses.[41]

10.2 Case Study 2: Bangladesh Arsenic Contamination — 60 Million People Exposed

Scale and impact. Bangladesh presents the world's largest documented mass poisoning from naturally occurring contamination. Approximately **60 million people** are exposed to arsenic concentrations above the WHO guideline of 10 μg/L from groundwater tube wells.[42][43] An estimated 20–30% of tube wells exceed safe limits. Arsenic causes bladder, lung, and skin cancers, peripheral vascular disease, and developmental impairment in children. The British Geological Survey's national assessment documented the geographic and hydrogeological drivers in detail; Harvey et al. (2002) in Science established the key mobilization mechanism.[44] The WHO has described this as one of the largest mass environmental poisoning events in human history.

The underlying challenge. Arsenic distribution in Bangladesh is spatially heterogeneous — variable at scales of hundreds of meters — and temporally dynamic. Arsenic mobilization is governed by redox conditions in the subsurface: in reducing environments, iron oxyhydroxide minerals dissolve, releasing adsorbed arsenic into groundwater. The critical drivers of spatial and temporal

[37] Natural Resources Defense Council, *Watered Down Justice*, NRDC, New York, 2019; and ongoing Flint water crisis documentation. https://www.nrdc.org/resources/whats-really-going-flint

[38] Masten, S.J., Davies, S.H. and McElmurry, S.P., *Flint Water Crisis: What Happened and Why?*, Journal AWWA, 108(12), 22–34, 2016; Edwards, M. et al., Virginia Tech Lead in Drinking Water Research, Virginia Tech, 2015–2019.

[39] Flint Water Advisory Task Force, *Final Report*, State of Michigan, March 2016. https://www.michigan.gov/documents/snyder/FWATF_FINAL_REPORT_21March2016_517805_7.pdf

[40] Masten, S.J., Davies, S.H. and McElmurry, S.P., *Flint Water Crisis: What Happened and Why?*, Journal AWWA, 108(12), 22–34, 2016; Edwards, M. et al., Virginia Tech Lead in Drinking Water Research, Virginia Tech, 2015–2019.

[41] Natural Resources Defense Council, *Watered Down Justice*, NRDC, New York, 2019; and ongoing Flint water crisis documentation. https://www.nrdc.org/resources/whats-really-going-flint

[42] World Bank, *Arsenic Contamination of Groundwater in South and East Asian Countries: Volume I — Policy Report*, World Bank, Washington, D.C., 2005. Report No. 31303.

[43] British Geological Survey / Mott MacDonald, *Arsenic Contamination of Groundwater in Bangladesh: Final Report*, British Geological Survey Technical Report WC/00/19, Keyworth, 2001.

[44] Harvey, C.F. et al., *Arsenic mobility and groundwater extraction in Bangladesh*, Science, 298(5598), 1602–1606, 2002.

variability include: groundwater depth and aquifer type; organic carbon content in sediments (which drives microbial reduction); seasonal water-table fluctuations linked to monsoon recharge and irrigation withdrawal; and longer-term water-table decline from intensified groundwater pumping for irrigation.[45][46]

Existing national-scale arsenic mapping (from the 2000 British Geological Survey survey and subsequent national testing programs) identifies which wells currently exceed safe limits. What it cannot predict is which wells will exceed safe limits under changing groundwater levels driven by climate change and irrigation expansion, or which currently borderline wells are trending toward unsafe concentrations. Static mapping cannot replace dynamic prediction.

τ-enabled change. A physics-faithful groundwater chemistry twin for Bangladesh's primary aquifer systems would model arsenic mobilization as a function of: seasonal water-table dynamics (measured or forecast); irrigation pumping patterns (known from national agricultural data); aquifer sediment geochemistry (from existing British Geological Survey and Bangladesh Water Development Board databases); and climate-projection inputs for changed monsoon recharge patterns. The outputs would include: spatially resolved arsenic-risk trajectories for wells under changing conditions; early warning for villages approaching deterioration thresholds; guidance for safe well placement and safe blending strategies between deep low-arsenic and shallow high-arsenic aquifers; and prioritized surveillance targeting for field testing programs operating under budget constraints.

The equity dimension is acute: the populations most dependent on at-risk tube wells are rural communities with limited access to alternative water sources and limited political power to demand responsive interventions. A prediction system that identifies emerging risk before exposure occurs is directly equitable.

Transferability. The arsenic-mobilization challenge has direct analogs in India (West Bengal, Uttar Pradesh, Bihar), Vietnam (Red River and Mekong deltas), Cambodia, Nepal, Pakistan, and parts of sub-Saharan Africa and Latin America. The same physics-faithful groundwater chemistry modeling capability applies wherever arsenic or fluoride contamination is driven by redox-sensitive geochemical processes under changing hydrogeological conditions.[47][48]

References. British Geological Survey Bangladesh arsenic assessment;[49] Harvey et al. (2002) in Science;[50] UNICEF Bangladesh WASH Programme documentation; World Bank Bangladesh Rural Water Supply and Sanitation Project assessments.[51]

11 ğ9 Finance, ROI, and Climate Finance

Global investment baseline. The development finance landscape for water safety is large and active. World Bank water and sanitation lending averages $3–4 billion per year, with specific programs including the Water Security and Climate Adaptation in Rural India (WASCA) program and the broader South Asia Water Initiative providing over $1 billion specifically for rural water

[45] Harvey, C.F. et al., *Arsenic mobility and groundwater extraction in Bangladesh*, Science, 298(5598), 1602–1606, 2002.

[46] Nickson, R. et al., *Mechanism of arsenic release to groundwater, Bangladesh and West Bengal*, Applied Geochemistry, 15(4), 403–413, 2000.

[47] Nickson, R. et al., *Mechanism of arsenic release to groundwater, Bangladesh and West Bengal*, Applied Geochemistry, 15(4), 403–413, 2000.

[48] Ravenscroft, P., Brammer, H. and Richards, K., *Arsenic Pollution: A Global Synthesis*, Wiley-Blackwell, Oxford, 2009.

[49] British Geological Survey / Mott MacDonald, *Arsenic Contamination of Groundwater in Bangladesh: Final Report*, British Geological Survey Technical Report WC/00/19, Keyworth, 2001.

[50] Harvey, C.F. et al., *Arsenic mobility and groundwater extraction in Bangladesh*, Science, 298(5598), 1602–1606, 2002.

[51] World Bank, *Arsenic Contamination of Groundwater in South and East Asian Countries: Volume I — Policy Report*, World Bank, Washington, D.C., 2005. Report No. 31303.

safety improvement.[52] UNICEF's WASH programme deploys over $800 million per year for water quality and WASH across developing countries.[53] The US EPA's Water Infrastructure Finance and Innovation Act (WIFIA) program provides low-interest loans for water infrastructure investment, with $5.5 billion in funding authorized.[54] The Green Climate Fund (GCF) has a freshwater access and climate adaptation window that has funded multiple water-quality-related climate adaptation programs.[55]

Deployment cost scenarios. Two illustrative cost scenarios ground the ROI discussion:

- *Scenario 1: City-scale source-water quality early warning (500,000–2 million people).* A τ source-water quality early warning system — covering source monitoring, model integration, treatment-plant decision support, and operator dashboards — is estimated at USD 2–8 million for initial deployment. Ongoing operational cost would be substantially lower. The capital cost is comparable to a few months of emergency tanker operation during a source contamination crisis, or to a small fraction of the remediation cost of a single major contamination event like Flint.
- *Scenario 2: National groundwater quality surveillance platform (Bangladesh model, 30–60 million people).* A national-scale arsenic-risk prediction and surveillance platform, integrated with existing monitoring programs and covering the primary affected aquifers, is estimated at USD 10–25 million. At 60 million people exposed, that represents a cost of USD 0.17–0.42 per person protected — among the most cost-effective public-health interventions available.

Benefit-to-cost ratios. The WHO/UNICEF 2012 analysis estimated that every $1 invested in water and sanitation yields $4.3 in economic returns, considering healthcare cost avoidance, productivity gains, and time-savings from reduced water collection burden.[56] Source-water quality early warning specifically — which reduces emergency treatment costs, prevents service interruptions, and avoids population-scale contamination events — is estimated to carry B:C ratios of **3:1 to 8:1** in the most relevant literature, depending on system size and baseline risk.[57]

Climate finance framing. Source-water quality early warning systems qualify for climate finance on two grounds: they are adaptation measures (improving resilience to climate-driven source-water deterioration) and they support both SDG 6 (water) and SDG 13 (climate action) simultaneously. The GCF, the Adaptation Fund, and bilateral climate finance channels (including USAID climate-resilience programs and the EU External Investment Plan) all have active water-sector windows where a τ translation pathway could be positioned.

Development bank appetite. The World Bank's Water Global Practice has articulated demand for improved water-quality intelligence tools in its water sector strategy documents.[58] The Asian Development Bank's water portfolio includes explicit calls for improved source-water monitoring and early warning in South and Southeast Asia — precisely the geography of the Bangladesh arsenic case study.[59] The African Development Bank's water security strategy calls for improved water-quality

[52] World Bank Water Global Practice, *Water Supply and Sanitation Overview*, World Bank, Washington, D.C., 2023. https://www.worldbank.org/en/topic/water/overview

[53] UNICEF, *WASH Annual Report 2023*, UNICEF, New York, 2023. https://www.unicef.org/reports/wash-annual-report-2023

[54] US Environmental Protection Agency, *Water Infrastructure Finance and Innovation Act (WIFIA) Program*, US EPA, Washington, D.C. https://www.epa.gov/wifia

[55] Green Climate Fund, *Water and Food Security Portfolio*, GCF, Incheon, 2023. https://www.greenclimate.fund/projects

[56] Hutton, G., *Global costs and benefits of drinking-water supply and sanitation interventions to reach the MDG M&E targets and universal coverage*, WHO/HSE/WSH/12.01, World Health Organization, Geneva, 2012.

[57] Hutton, G. and Chase, C., *The knowledge base for achieving the sustainable development goal targets on water supply, sanitation and hygiene*, International Journal of Environmental Research and Public Health, 13(6), 536, 2016.

[58] World Bank Water Global Practice, *Water Supply and Sanitation Overview*, World Bank, Washington, D.C., 2023. https://www.worldbank.org/en/topic/water/overview

[59] Asian Development Bank, *Water Security for Asia: Strategic Issues and Directions*, ADB, Manila, 2022. https://www.adb.org/publications/water-security-asia

surveillance and early warning across sub-Saharan Africa.[60] All three represent active procurement and project-development channels.

12 ğ10 Evidence and Translation Ladder

The deployment ladder for Paper 1 is organized around four phases that progress from shadow-mode validation through full climate-resilient integration. Each phase has explicit deliverables and goals that build on the preceding phase.

Phase 1 — Shadow-Mode Source-to-Works Intelligence (Months 1–18). Deploy τ in parallel with existing utility practice at one or two lighthouse pilots. The system makes predictions but does not yet influence operations. Deliverables: source-water hazard prediction for intake conditions (turbidity, bloom risk, contamination arrival); treatment stress forecasting linked to source predictions; post-event reconstruction and prediction verification; operator-facing dashboards presenting predictions alongside current sensor readings. Goal: demonstrate predictive superiority over current practice in blinded comparison without operational risk. The shadow-mode phase is the evidence-generation phase.

Phase 2 — Integrated Treatment Optimization Support (Months 12–36). Connect the validated source-risk layer to treatment plant operations. Deliverables: predictive chemical dosing support (coagulant, disinfectant, pH adjustment); filtration and membrane management scenario support; contingency-state transition recommendations with explicit confidence intervals; energy and chemical consumption optimization linked to source forecasts. Goal: reduce avoidable operator uncertainty and demonstrate measurable improvement in treatment efficiency and water quality consistency.

Phase 3 — Climate-Resilient Water Safety Plan Execution Layer (Months 24–48). Embed τ in utility WSP and climate-resilient WSP workflows. Deliverables: executable risk scenarios for compound hazards (flood + contamination, drought + concentration, heat + power stress); stress-test libraries for regulatory submission; adaptation-investment priority maps derived from quantified scenario testing; regulator-facing resilience evidence packages. Goal: transform climate-resilient water safety planning from qualitative narrative into a living operational twin with explicit, testable risk quantification.

Phase 4 — Equity-Focused Rollout to Smaller and Fragile Systems (Months 36–60). Adapt the platform for small utilities, rural systems, decentralized treatment, and climate-fragile geographies. Deliverables: coarse-grained τ packages requiring minimal sensor infrastructure; simplified decision-support outputs calibrated for lower-capacity operators; mobile-compatible alerting for rural water operators; integration with national surveillance programs (JMP, GLASS, national regulators). Goal: extend public-good reach beyond large utilities to the highest-equity populations, demonstrating that the same physical model can operate across a full spectrum of institutional capacity.

13 ğ11 Stakeholder Map

Effective deployment of source-to-works intelligence requires coordinated engagement with stakeholders across five functional roles.

Water utilities (large and small). Primary operational deployers. Large utilities — metropolitan authorities, national water companies in middle-income countries — are the first-mover target and the lighthouse pilot environment. Small utilities and rural water operators are the highest-equity

[60] African Development Bank, *African Water Vision 2025 — Equitable and Sustainable Use of Water for Socioeconomic Development*, AfDB and AMCOW, Abidjan, 2012; AfDB Water Sector Strategy 2021–2025.

target and require adapted interfaces. Key institutional champions: utility operations directors, water quality managers, and plant engineers who bear direct accountability for treatment performance.

National drinking-water regulators and environment ministries. Set compliance standards, approve treatment processes, and hold utilities accountable for water quality. A shared regulatory view into the source-to-works twin strengthens both compliance monitoring and emergency management. Key champions: drinking-water inspectorates, environment agency water quality divisions, and national public health laboratory networks.

Health ministries and national public health institutes. Own the disease burden and surveillance mandate. WHO burden-of-disease data on unsafe WASH are produced by or in partnership with national health ministries. When source-to-works intelligence prevents a contamination event, the health ministry sees the averted disease burden in its surveillance data. Key champions: epidemiology units, environmental health departments, and emergency health operations centers.

Development banks and climate-finance institutions. World Bank, Asian Development Bank, African Development Bank, Green Climate Fund, and bilateral development finance institutions are the primary capital providers for water infrastructure in low- and middle-income countries. Their project cycles and procurement mechanisms are the primary entry channel for large-scale τ deployment in developing-country utilities. Key champions: water sector specialists, climate adaptation portfolio managers, and monitoring-and-evaluation units that need outcome metrics for water safety investments.

WHO, UNICEF, UNEP, and UN-Water. Custodians of the normative frameworks (WSP, JMP, GEMS/Water, SDG 6 indicators) that define the mission τ is addressing. Engagement with these agencies serves two functions: normative alignment (ensuring τ deployments are framed within WHO WSP methodology) and market development (positioning τ early warning as an implementation tool for official SDG 6 commitments). Key champions: WHO water quality and health team, UNICEF WASH programming, UNEP GEMS/Water secretariat.

Research institutions and technology transfer hubs. University water research groups (Virginia Tech, Delft IHE, EAWAG, IIT Delhi) and international water research institutes (IWMI, IRC) provide technical validation, pilot partnership, and technology adaptation capacity. Marc Edwards' group at Virginia Tech — which documented the Flint crisis and the underlying corrosion chemistry — is an exemplary model of research-to-operations engagement that τ deployment should emulate.[61]

14 ğ12 Gender, Equity, and Labor Dimensions

Gender and time poverty. In low- and middle-income countries, water collection is disproportionately a female and child responsibility. WHO and UNICEF data consistently show that women and girls spend an average of 6 hours per day on water collection in households lacking on-premises service.[62] Water quality failures — contamination events, system shutdowns, unsafe source conditions — extend collection burdens or force unsafe source substitution. Source-to-works early warning that reduces service interruptions and contamination events has a direct gender dividend: time returned to women and girls for education, economic activity, and health.

Child health equity. Children under five bear the highest diarrheal disease burden attributable to unsafe water. The Flint crisis documented the developmental consequences of lead exposure in children acutely. Bangladesh's arsenic burden similarly concentrates developmental harm in children during critical growth windows. Source-water quality early warning is disproportionately a child

[61] Masten, S.J., Davies, S.H. and McElmurry, S.P., *Flint Water Crisis: What Happened and Why?*, Journal AWWA, 108(12), 22–34, 2016; Edwards, M. et al., Virginia Tech Lead in Drinking Water Research, Virginia Tech, 2015–2019.

[62] WHO/UNICEF, *Progress on drinking water, sanitation and hygiene: 2017 update and SDG baselines*, WHO and UNICEF, Geneva, 2017, p. 26. https://www.who.int/publications/i/item/9789241512893

health intervention.

Small-system equity. The populations served by small and rural water systems are typically poorer, have fewer alternatives, and receive less sophisticated technical oversight than urban utility customers. Deployment strategies that explicitly target small-system adaptation — Phase 4 of the deployment ladder — are not optional equity additions; they are core to the public-good case for τ in this domain. A technology that improves water safety only for large, well-resourced utilities while leaving rural populations unserved does not address the SDG 6 access gap.

Labor and operator capacity. Water utility operators in low- and middle-income countries often work under significant informational stress: they make treatment decisions with inadequate source-water data, under time pressure, with limited technical backup. A decision-support system that reduces operator uncertainty — particularly during unusual events — is directly a labor welfare intervention. The goal is not to replace operators but to give them better information under challenging conditions, reducing the burden of making consequential safety decisions on incomplete data.

Climate vulnerability intersection. The populations most exposed to climate-driven source-water deterioration — coastal communities facing saltwater intrusion, dryland communities facing drought concentration, deltaic communities facing arsenic mobilization from changing water tables — are frequently also the most economically vulnerable. Climate finance channels that direct τ deployment to climate-vulnerable geographies accordingly address multiple equity dimensions simultaneously.

15 ğ13 Benchmark Suite

A rigorous benchmark suite for Paper 1 should test τ's source-to-works capabilities across six operational challenge classes. Success criteria are defined not only by predictive accuracy but by operational utility: the prediction must arrive with sufficient lead time to be actionable, with bounds narrow enough to support treatment decisions.

Benchmark 1 — Storm-driven turbidity event prediction. Predict intake turbidity spikes and peak timing following major precipitation events. Success criteria: arrival timing accurate within ± 2 hours; peak turbidity within $\pm 15\%$ of observed; bounded uncertainty interval correctly covering observed outcome in $\geq 90\%$ of events. Baseline comparison: current utility practice (typically reactive detection 0–2 hours before arrival).

Benchmark 2 — Flood-driven contamination risk prediction. Predict when flood events or combined sewer overflow conditions threaten microbial or chemical contamination of surface water sources. Success criteria: detection of high-risk episodes with ≥ 6 hours lead time; specificity sufficient to avoid excessive false positives that would trigger unnecessary treatment escalation. Baseline comparison: current risk advisory systems (threshold-triggered, mostly reactive).

Benchmark 3 — Salinity intrusion early warning. Predict when salinity intrusion will breach safe thresholds at coastal, estuarine, and deltaic intakes, particularly under climate-driven sea-level and drought conditions. Success criteria: ≥ 24 hours warning before threshold breach; accurate prediction of intrusion depth and duration. Baseline comparison: tide-gauge and conductivity sensor monitoring (typically 0–4 hours warning).

Benchmark 4 — Cyanobacterial bloom and reservoir intake stress. Predict operationally relevant water-quality deterioration from cyanobacterial bloom development and related reservoir stratification. Success criteria: bloom onset prediction ≥ 3 days before operational impact; toxin-producing genus differentiation where data support it. Baseline comparison: satellite-based bloom detection and manual sampling (typically reactive).

Benchmark 5 — Treatment-train response fidelity. Given a source-water quality forecast, predict what that forecast implies for coagulant demand, disinfection by-product precursor load,

filtration run length, and membrane differential pressure. Success criteria: coagulant demand prediction within ±10% of observed; DBP formation potential prediction within ±15%; filtration runtime prediction within ±20%. Baseline comparison: empirical correlations and operator experience.

Benchmark 6 — Climate-resilient WSP compound-event stress testing. Generate quantitative safety-margin estimates for compound hazard scenarios (drought + concentration, flood + contamination, heat + power + demand stress) that can be validated against historical extreme events and accepted as credible by regulators for climate-resilient WSP submissions. Success criteria: scenario outputs accepted by a national drinking-water regulator as technically credible inputs to an official climate-resilient WSP. Baseline comparison: qualitative risk matrix assessments.

16 ğ14 Governance Guardrails

Responsible deployment of source-to-works intelligence requires explicit governance commitments. The following five guardrails define the institutional contract between τ as a tool and the water sector as a public-interest domain.

Guardrail 1 — Utility and regulator oversight remains primary. The τ layer provides decision support, not autonomous decision-making. Operational decisions — treatment adjustments, source switching, service advisories, emergency declarations — remain the responsibility of licensed water professionals and competent regulatory authorities. τ augments professional judgment; it does not replace it or bypass public accountability.

Guardrail 2 — Auditability and explainability are non-negotiable. Drinking-water safety decisions must be defensible in regulatory proceedings, public inquiries, and — in the worst cases — legal proceedings. A black-box prediction system that cannot explain its outputs in terms a water engineer can interrogate is not acceptable in this domain. Every τ prediction must be accompanied by an explicit physical explanation of its basis and a traceable uncertainty quantification.

Guardrail 3 — Equity must be built in, not added on. Translation pathways must allocate explicit resources for small-utility and rural-system adaptation from the outset. Phase 4 (equity-focused rollout) is not contingent on Phase 1–3 commercial success; it is a design requirement for programs funded by development banks or climate-finance institutions operating under equity mandates.

Guardrail 4 — Fail-safe design is a first-order requirement. Water operators care not only about average performance but about what happens during rare, bad, compound events — exactly when cognitive load is highest and consequences of error are most severe. The τ system must degrade gracefully: when data inputs are missing or unreliable, predictions must widen their uncertainty bounds and flag reduced confidence rather than returning false precision. Operators must trust that the system knows what it does not know.

Guardrail 5 — Source protection and treatment intelligence must not be institutionally separated. If τ is deployed only at the treatment plant and not at the catchment and source level, most of its value for drinking-water safety is lost. The catchment-to-consumer logic is not a design aspiration; it is an institutional requirement. Governance arrangements must ensure that source-water monitoring, treatment operations, and distribution-network management access a shared intelligence layer rather than operating in institutional silos.

17 ğ15 SDG Mapping and Bottom Line

SDG 6 — Clean Water and Sanitation. Paper 1 addresses SDG 6.1 (universal safe and affordable drinking water), SDG 6.3 (improved water quality and ambient water quality surveillance), and SDG 6.b (participation and integrated water resources management). The τ source-to-works

twin is a direct operational tool for SDG 6.1 progress: it makes "safely managed" — the hardest tier of the SDG 6.1 ladder — more achievable for systems currently delivering only "improved" or "basic" service.

SDG 3 — Good Health and Well-Being. Source-water quality early warning directly attacks SDG 3.2 (end preventable deaths of children under five), SDG 3.3 (end epidemic communicable diseases including cholera and typhoid), and SDG 3.9 (reduce illness and death from water contamination). The 14,000–42,000 avoidable deaths per year estimate from ğ5 maps directly onto SDG 3 progress.

SDG 13 — Climate Action. Climate-resilient water safety planning (Opportunity Area C, Phase 3) is a direct climate adaptation measure under SDG 13.1 (strengthen resilience and adaptive capacity to climate hazards) and SDG 13.b (promote climate change planning in LDCs and SIDS). Source-water systems are among the most directly climate-exposed public health infrastructure in most countries.

SDG 10 — Reduced Inequalities. The equity-focused rollout (Phase 4) and the rural/small-system priority in the opportunity map address SDG 10 by ensuring that source-to-works intelligence benefits do not accrue only to large, well-resourced utilities serving wealthier urban populations. The Bangladesh case study illustrates the intersection: the 60 million people most acutely exposed to arsenic poisoning are among the poorest and most politically marginalized populations in South Asia.

Bottom Line. Under the working τ assumption, drinking-water source, treatment, and quality early warning is one of the most compelling public-good entry points in the entire broader τ impact portfolio. The official sector has already defined the mission: catchment-to-consumer risk management, water quality early warning, and climate-resilient water safety planning. What the official sector lacks is the physically faithful, bounded-error, predictive intelligence layer that would make that mission executable at utility operations tempo.

τ claims to provide exactly that. If the claim is sound, the consequences are:

- safer drinking water for millions of people currently served by systems operating at or near the edge of their treatment capacity;
- fewer avoidable waterborne disease deaths and disability-adjusted life years lost;
- stronger climate resilience in a domain where climate change is rapidly narrowing operating margins;
- better regulatory governance with fewer blind spots between catchment and tap;
- and a more equitable water future in which analytical protection extends to the smallest and most vulnerable systems, not only to the largest and best-resourced.

That is why Paper 1 comes first.

18 References

Source: Full manuscript text integrated from Public-Good Briefing draft.

19 Dossier accountability addendum

The following addendum records the release-facing accountability layer for this dossier: claim boundaries, baseline evidence, upstream dependencies, translation assumptions, scenario bands, scorecard rationales, benchmark requirements, governance guardrails, and related Panta Rhei surfaces. It is intentionally downstream of the full source argument above.

19.1 Executive summary

A Public-Good Briefing showing how a law-faithful tau drinking-water twin could unlock major public-good gains in source-water early warning, treatment-train optimization, and climate-resilient water-safety planning—addressing 2.1 billion people lacking safely managed drinking water.

Impact thesis

A Public-Good Briefing showing how a law-faithful tau drinking-water twin could unlock major public-good gains in source-water early warning, treatment-train optimization, and climate-resilient water-safety planning—addressing 2.1 billion people lacking safely managed drinking water. The v3 impact thesis is conditional: a Tau-grade drinking-water source, treatment, quality, and early-warning twin would become valuable if it improves benchmarked public decisions while preserving transparent uncertainty, reviewability, and governance control.

19.2 Public-good burden and baseline evidence

A Public-Good Briefing showing how a law-faithful tau drinking-water twin could unlock major public-good gains in source-water early warning, treatment-train optimization, and climate-resilient water-safety planning—addressing 2.1 billion people lacking safely managed drinking water. The public-good burden is treated here as an institutional decision problem: existing agencies already monitor parts of the domain, but the operational handoff from data to timely, auditable action remains incomplete.

19.2.1 External evidence baseline

- **WHO/UNICEF Joint Monitoring Programme**, Progress on household drinking water, sanitation and hygiene 2000–2022: special focus on gender, WHO and UNICEF, Geneva, 2023 [12]: source-page evidence item.
- **WHO news release**, 1 in 4 people globally still lack access to safe drinking water, 26 August 2025 [11]: source-page evidence item.
- **WHO**, Water, sanitation and hygiene: burden of disease, Global Health Observatory [10]: source-page evidence item.
- **UNESCO / UN-Water**, UN World Water Development Report 2024: Valuing Water [7]: Statistics, UNESCO, Paris, 2024.
- **UN-Water**, Progress on Wastewater Treatment – 2024 Update, UN-Water, Geneva, 2024 [4]: source-page evidence item.
- **UN-Water**, Progress on Water-related Ecosystems – 2024 Update, UN-Water, Geneva, 2024 [5]: source-page evidence item.
- **WHO**, Water safety planning, WHO water, sanitation and health [9]: source-page evidence item.

- **WHO**, Climate-resilient water safety plans: managing health risks associated with climate variability and change, WHO, Geneva, 2017 [8]: source-page evidence item.
- **UNEP GEMS/Water**, SDG 6.3.2 Indicator: Ambient Water Quality [6]: source-page evidence item.
- **UN-Water**, Progress on Ambient Water Quality – 2024 Update, SDG Indicator 6.3.2 Report, UN-Water, Geneva, 2024 [3]: source-page evidence item.

19.3 Current institutional landscape

The relevant landscape includes public agencies, research infrastructures, standards bodies, development-finance channels, and domain review communities represented in the evidence base, including UN-Water, UNESCO / UN-Water, WHO, WHO news release, WHO/UNICEF Joint Monitoring Programme. These references are evidence and adoption surfaces, not endorsements or deployment partners.

19.4 Capability gap

The practical gap is a benchmarkable translation gap: current systems expose useful data or partial models, but they do not yet provide a single law-faithful, bounded-error decision layer for drinking-water source, treatment, quality, and early-warning twin.

19.5 Tau framework dependency map

Surface	Role in this dossier
Build the Tau-Kernel	finite address and scalar foundation
Recover Core Mathematics	mathematical bridge and model interface
Derive Physics	physical readout and domain translation candidate
Results lane	upstream consequences to be mapped precisely during release preparation
direct-registry-mapping-withheld	no direct Registry object is asserted until a substantive Corpus mapping is available
public-docs-mapping-withheld	TauLib module links are asserted only where public documentation exposes a clear surface
Release Manifest	release baseline
Predictions and Falsification	empirical accountability route

19.6 Translation assumptions and missing engineering

Required domain model: **drinking-water source, treatment, quality, and early-warning twin**.

First benchmarkable test: source-water risk, treatment response, and quality-event warning against monitoring, treatment, and incident records.

- domain-specific model construction
- data ingestion and validation
- benchmark harness
- pilot protocol

- independent review workflow

19.7 Impact mechanism chain

Public-good burden → external evidence baseline → τ capability hypothesis → upstream Results / Corpus / Verify dependency → translation assumptions → benchmarked pilot → governed adoption pathway.

19.8 Scenario bands

Band	Scenario summary	Confidence
Conservative	A narrow shadow-mode pilot improves one bounded decision task for Drinking-Water Source, Treatment, and Quality Early Warning without operational authority.	medium
Realistic	A reviewed prototype strengthens several public-sector workflows for Drinking-Water Source, Treatment, and Quality Early Warning after benchmark comparison with incumbent systems.	medium-low
Optimistic	A reusable public-good intelligence layer becomes plausible for Drinking-Water Source, Treatment, and Quality Early Warning after external validation and transparent governance review.	low

19.9 Impact scorecard

Public-good scale	5/5	The affected public-good burden is large or institutionally significant within the portfolio.
Tau fit	4/5	The proposed pathway depends on coupled state, bounded uncertainty, and compositional modelling rather than isolated prediction alone.
Evidence proximity	5/5	The evidence base is anchored in public institutions, official monitoring systems, or established scientific reviews.
Measurability	5/5	A first benchmark can be framed against incumbent public datasets, institutional records, or operational decision metrics.
Adoption readiness	3/5	Adoption remains conditional on domain review, governance fit, data access, and institutional integration.
Equity leverage	5/5	The pathway can prioritize underserved or vulnerable populations where public access and safeguards are built in.

19.10 Candidate pilot pathways

water-safety early-warning pilot with a public utility and health/environment regulator

19.11 Benchmark suite and success metrics

Type	Incumbent baseline	Required Tau benchmark	Success metric	Validator
translation benchmark	current public or institutional systems in the domain	source-water risk, treatment response, and quality-event warning against monitoring, treatment, and incident records	pre-registered accuracy, latency, uncertainty, or decision-quality metric	independent domain reviewers
governance benchmark	existing audit, disclosure, and reporting practice	transparent assumption, data, model, and failure-mode disclosure	reviewable evidence pack and adverse-outcome protocol	public-sector or expert governance panel
equity benchmark	current access, service-quality, or exposure disparities	documented pathway for underserved or vulnerable users without hidden exclusion	distributional benefit and risk review before pilot expansion	equity, community, or public-interest review process

19.12 Governance and risk guardrails

- Human oversight for any operational use.
- Public benchmark disclosure before institutional adoption.
- Equity access review for underserved or vulnerable communities.
- Data-rights and privacy controls for operational datasets.
- Misuse-prevention and adverse-outcome monitoring.
- Adverse-outcome monitoring with a documented escalation path.
- External domain review before pilot expansion.

19.13 Related Results / Corpus / Verify / Publications

This dossier is downstream of Results, Corpus, Verify, and Publications surfaces. It is not a Registry object. Direct Registry or TauLib links are asserted only where the mapping is substantive rather than decorative.

19.14 Bibliography and external evidence

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